



It's Only a Model:

Demystifying
Row Level Security
in Power BI

Leslie Welch
Principal Architect, Data & AI
SHI International





(adult) children



2

household



4

garden beds



9

years in data viz



11

years climbing



14

obstacle races



16

years coding



33

let

```
OffTime = Table.AddColumn(Workweek_Is_Over, Hobbies, each if  
[Weather]="Nice" then [ObstacleRace] else [IndoorClimbing])  
in OffTime
```



/leslie-welch



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www.powerbidc.org



RLS Workshop Scope

- Authentication vs. Access vs. Row Level Security (RLS)
- RLS Methods
- Performance Impact and Considerations
- Modeling for RLS
- RLS use cases
- Considerations with cross-tenant collaboration
- Considerations with cross-cloud collaboration

×

×

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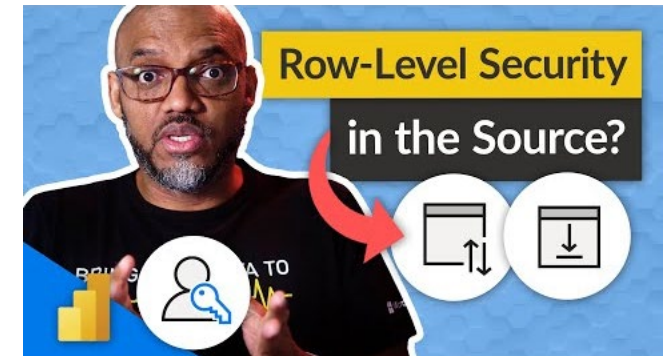
× Out of scope for this talk



<https://github.com/lesliewelch/powerbi>

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 - ✗ • RLS with composite models
 - ✗ • Object Level Security (OLS)
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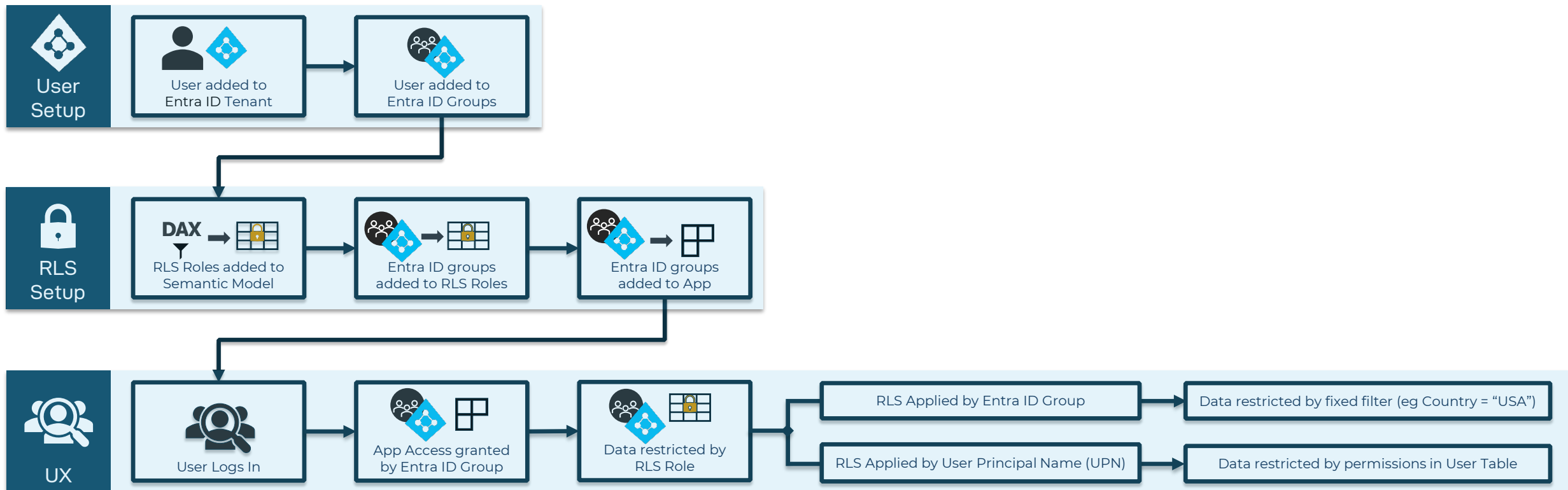
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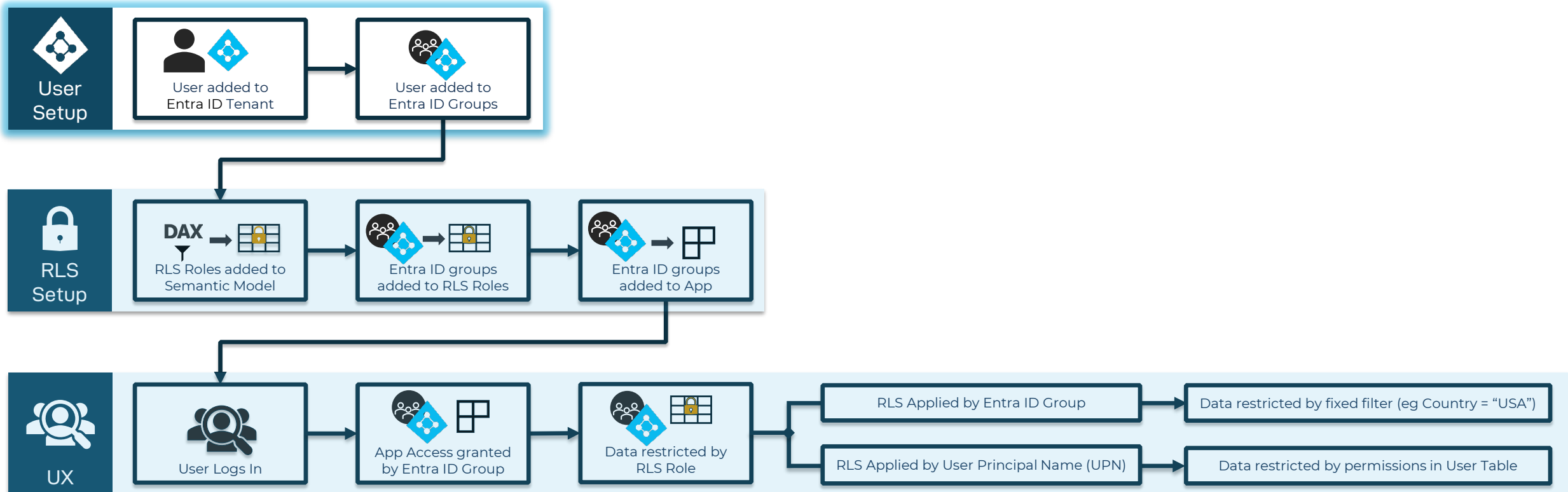
Authentication
and Access
and RLS, oh my!

NONE SHALL PASS

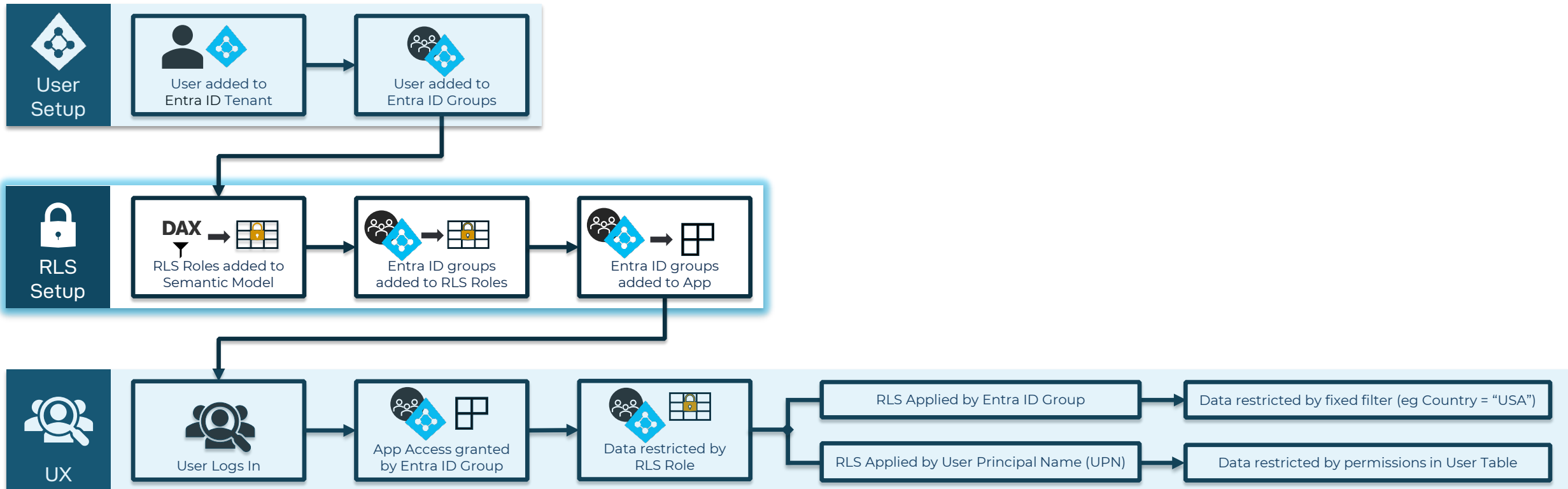
Access + RLS Overview – User Flow



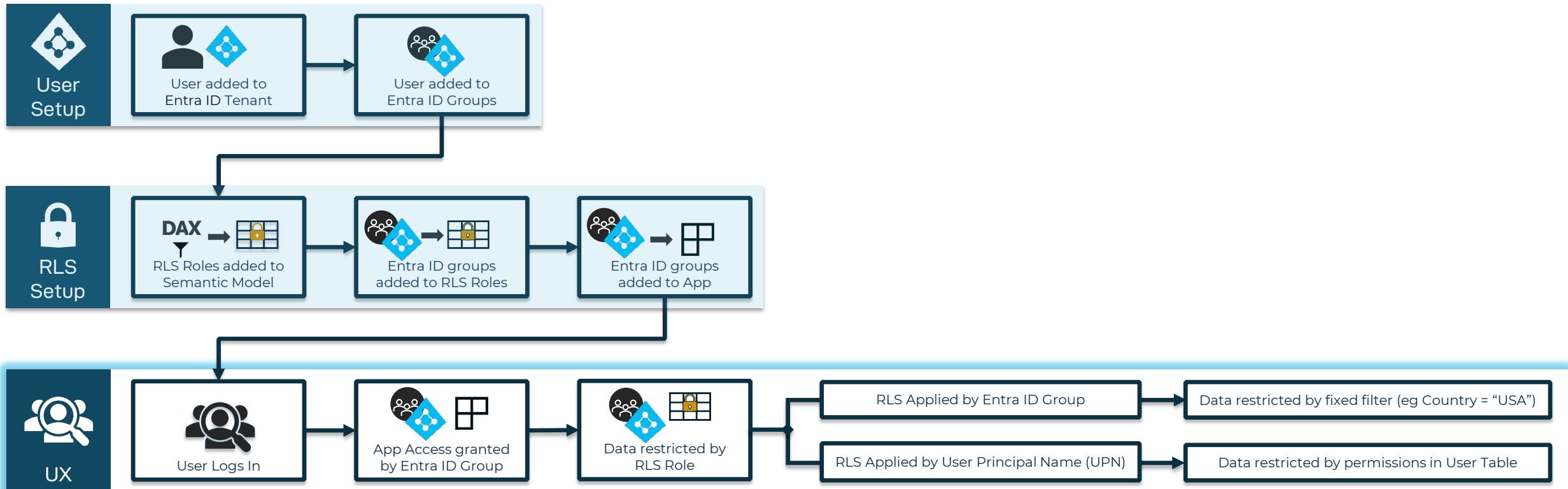
Access + RLS Overview – User Flow



Access + RLS Overview – User Flow

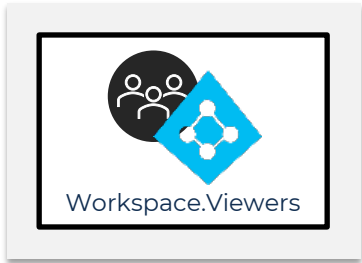


Access + RLS Overview – User Flow



Access + RLS Overview

Access

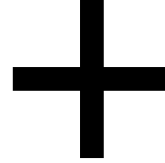


Individual Users are assigned to one or more groups for **access** to one or more reports, apps, or workspaces.



Gives **access** to **structure** / framework of App, Report, Dashboard, Scorecard, or Semantic Model.

Does not give access to any data if Row Level Security is in place within Power BI Semantic Model.



Row Level Security (RLS)



Individual users are assigned to one or more groups for **RLS**.

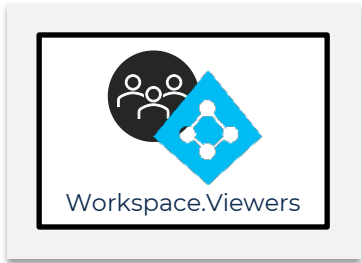
2	Monday, December 28, 2026	2048
2	Saturday, December 19, 2026	2050
2	Wednesday, December 9, 2026	2014
2	Wednesday, December 2, 2026	2022

Gives **access** to record level **data** aligned with filter logic in RLS Security Role for the underlying Semantic Model.

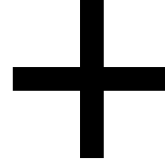
Does not give access to structure/container that contains the data
(App, Report, Workspace, Semantic Model etc.)

Access + RLS Overview

Access



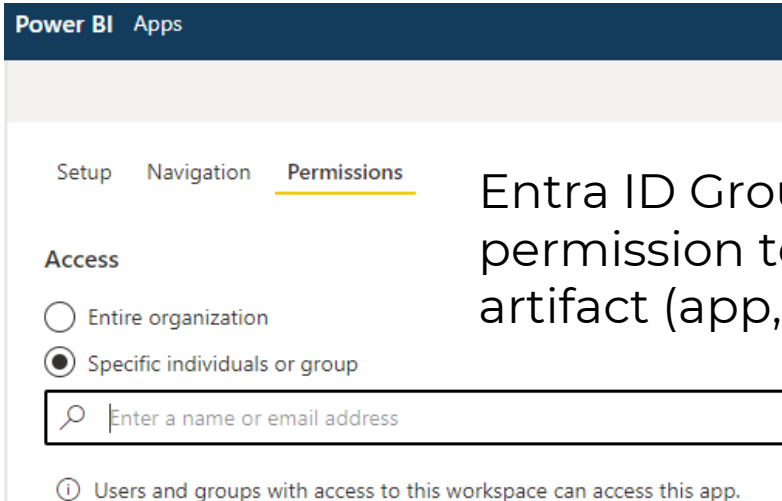
Individual Users are assigned to one or more groups for **access** to one or more reports, apps, or workspaces.



Row Level Security (RLS)

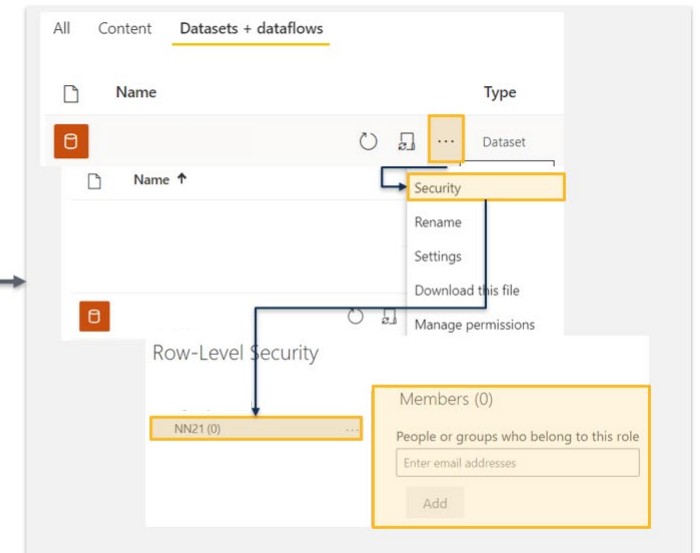
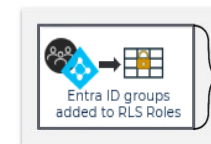


Individual users are assigned to one or more groups for **RLS**.



Entra ID Groups granted permission to Power BI artifact (app, report, et al.)

Entra ID Groups assigned to RLS Security Role



Access + RLS Overview

Access to App/Report	Access to Semantic Model	Assigned to Security Role	UPN exists in User Table	
✓	✓	✗	✓	✗ Error shows RLS applied, contact owner
✓	✓	✓	✗	✗ Error shows RLS applied, contact owner
✓	✗	✓	✓	✗ No error, no way to access
✗	✓	✓	✓	✗ No error, no way to access
✓	✓	✓	✓	✓ Access app/report and data

Sorry, we couldn't find that report ✕

It may have been deleted or you may not have access to view it.

[Get help](#)

OK

Access denied

You can't see the content of this report because you don't have permissions to the underlying dataset. The underlying dataset uses row-level security (RLS). Contact the dataset owner

to view this report.

Close

RLS Methods



RLS Methods & Considerations

Dynamic - User Principal Name (UPN)

1



Department ID	Organization ID	UPN
2001	1001	emma.adams@ConfuseACatLtd.co.uk
2002	1001	noah.johnson@ConfuseACatLtd.co.uk
2003	1001	olivia.thompson@ConfuseACatLtd.co.uk
2004	1001	liam.williams@ConfuseACatLtd.co.uk
2005	1001	ava.brown@ConfuseACatLtd.co.uk
2006	1001	ethan.davis@ConfuseACatLtd.co.uk
2007	1001	sophia.white@ConfuseACatLtd.co.uk

Permissions defined by User Table added to model.

One RLS Role for all users: `UPN = USERPRINCIPALNAME()`

- ⚠ More RLS DAX evaluation (due to granularity)
- ⚠ Larger Model (due to User Table)
- ⚠ DAX measures may be more complex to support visuals

Static - EntraID Group (fixed access per group)

*



*



Department ID	Organization ID
2001	1001
2002	1001
2003	1001
2004	1001
2005	1001
2006	1001
2007	1001

Permissions defined by Entra ID Group.
No User Table needed.

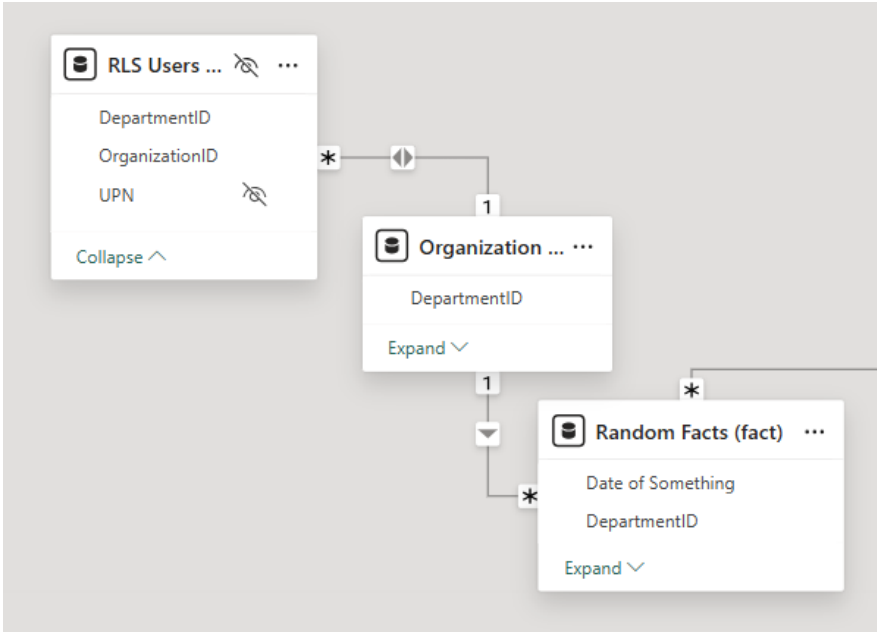
One Entra ID group for each fixed access permission.

- ⚠ More Entra ID Security Groups
- ⚠ More Power BI Security Roles
- ⚠ Automating Role creation and editing requires using Tabular Editor; once used model cannot be edited in Power BI Desktop

*maybe

Dynamic RLS

Example shows User Table with Organization and Department values associated with users.



DAX for Example RLS Role:

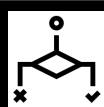
[UPN] = USERPRINCIPALNAME()

Evaluate each row of 'RLS Users' and filter out False

User ethan.davis@ConfuseACatLtd.co.uk logs in

DAX function USERPRINCIPALNAME() returns email address – ethan.davis@ConfuseACatLtd.co.uk

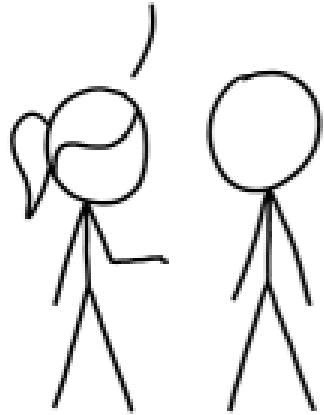
User Table filtered to relevant record(s), This filters 'Organization (dimension for Dynamic RLS)' table, which in turn filters the fact table.

 What Reports should I see?	 What logic should apply to me?	 What rows should be returned for me?
Entra ID Groups	Entra ID Groups	User Table
Audiences (Reports & other artifacts can be shown or hidden per audience)	Power BI Security Roles (set up in Desktop, Entra ID group assigned to Role in published Semantic Model)	User Table is filtered based on email for Agency role, Gov-wide role filters all records except gov-wide aggregate, Support filters no records

Row Level Security Methods

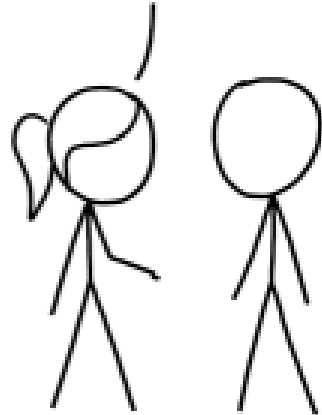


IT TOOK SOME EXTRA
WORK TO BUILD, BUT
NOW WE'LL BE ABLE
TO USE IT FOR ALL OUR
FUTURE PROJECTS.



HOW TO ENSURE YOUR
CODE IS NEVER REUSED

LET'S NOT OVERTHINK IT;
IF THIS CODE IS STILL IN
USE *THAT* FAR IN THE
FUTURE, WE'LL HAVE
BIGGER PROBLEMS.



HOW TO ENSURE YOUR
CODE LIVES FOREVER

Performance Considerations & Modeling for RLS

Data Modeling & Performance



Data Source



Model Schema



Data Security



Model Size



Aggregation
Methods

“

★ A bidirectional cross-filter is going to quickly create our worst nightmare, because it introduces ambiguity in the model.

Alberto Ferrari, SQLBI co-founder and top author for DAX content



“



After all, if you need to kill ants in the yard,
you do not turn on the Death Star.

Alberto Ferrari, SQLBI co-founder and top author for DAX content



Data Modeling – Star Schema

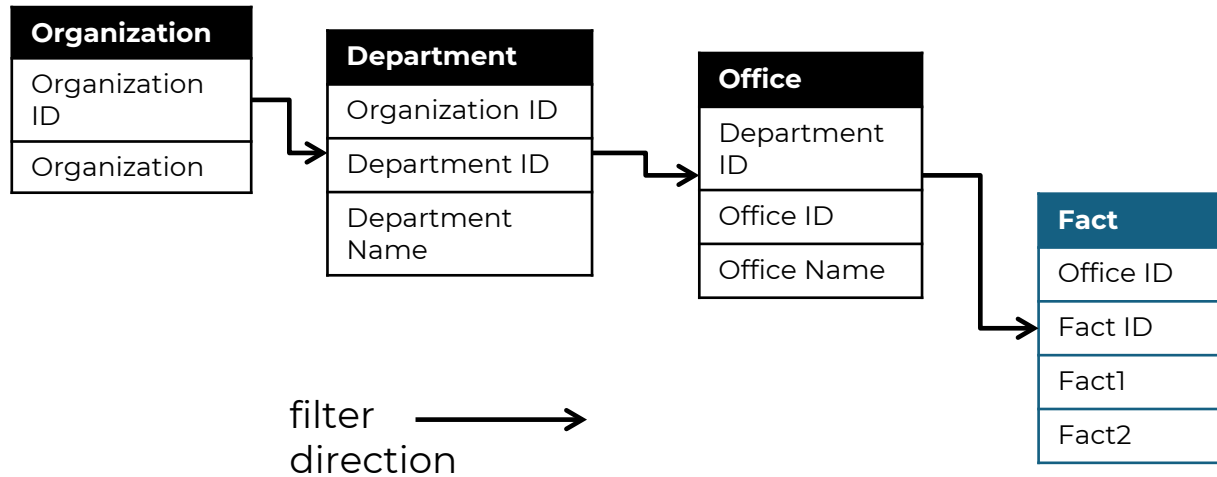


<https://youtube.com/clip/UgkxYJvme54Olhazh3vOnAMy6kei3G-FQznO>

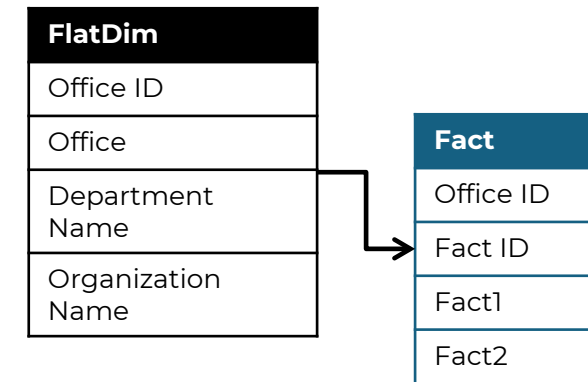
Data Modeling – Star Schema



snowflake schema



star schema



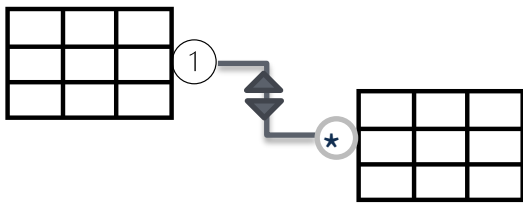
“



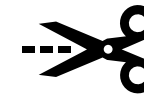
Even though a star schema might require a few ETL steps and some coding, it is *always* the best modeling option.

Alberto Ferrari, SQLBI co-founder and top author for DAX content

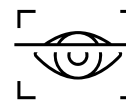
Data Modeling – Performance



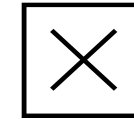
Loaded & visible:
Fields used in
visuals or slicers.



Removed in Power
Query:
Fields used for PQ
calculations.



Loaded & hidden:
Fields used in
relationships or
DAX measures
only.



Not brought into
Power Query:
Fields not used at all.

**Avoid bi-directional
relationships.**

**Make record-level
transformations in data
source or Power Query.**

Only bring in what you need!



Tips from the pros: “Roche’s Maxim of Data Transformation: Data should be transformed as upstream as possible, and as far downstream as necessary.”
Matthew Roche, Fabric CAT Team

Do I even need RLS?

4.5M

avg. cost per
breach



Pro-tip:



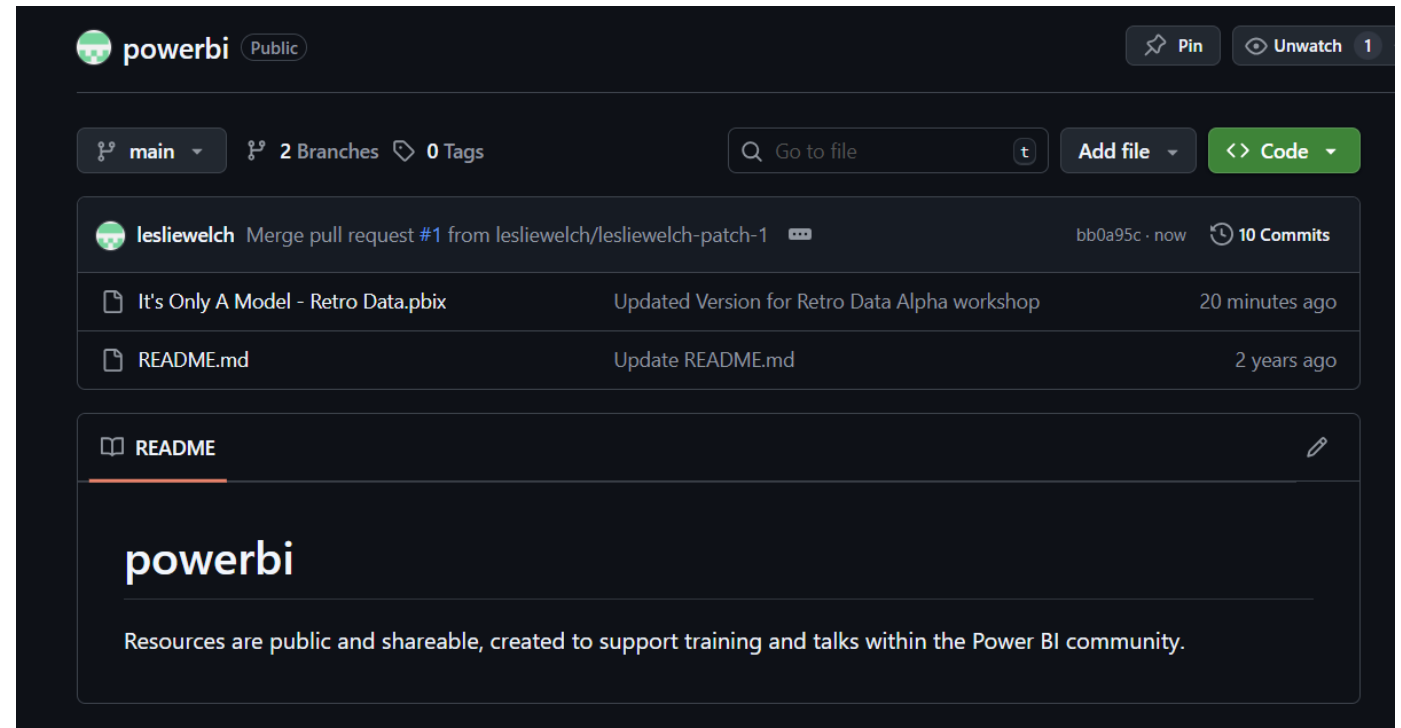
**Don't
Procrastinate!**



**GOD KNOWS I'D LOVE TO INCORPORATE ROW LEVEL
SECURITY IN A POWER BI MODEL AFTER BUILDING IT**

Let's Dig In: Setting up Row Level Security

<https://github.com/lesliewelch/powerbi>



Learn more

[Guy In A Cube - RLS in the source?](#)

[Guy In A Cube - Row-Level Security and Object-Level Security together in Power BI??? Yup!](#)

[RADACAD - RLS on a DirectQuery to Power BI dataset composite model My Findings](#)

[SQLBI - Header-detail vs star schema models in Tabular and Power BI](#)

[SQLBI - Bidirectional relationships and ambiguity](#)

[SQLBI - Bidirectional relationships and ambiguity in DAX \(blog\)](#)